

A Minimal Axiom System for Causal Composition: Forced Collapse, Finite-Free Dichotomy, and a Unique Prime Basis

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Abstract

We present a minimal axiom system for causal composition, formalized in Lean 4 with Mathlib. The system consists of four axioms: a total binary composition operation, its associativity, an exact additivity law for dependency lists, and a no-duplicate constraint. From these four axioms we prove *Core Collapse*: every rule has an empty dependency list. We further prove that associativity is redundant for this result, and that the exact additivity law cannot be weakened to sub-additivity. We then show that the sequence $s : \mathbb{N} \rightarrow \mathbb{N}$ with $s_0 = 2$, $s_{n+1} = 3 + 2n$ generates the prime basis $\{2, 3, 5, 7\}$ of three finite simple groups A_4 , A_5 , and $\text{PSL}(2, 7)$, and that this basis is uniquely determined by an exclusion argument. All results are machine-verified (2089 jobs, 0 errors, 0 sorry) across 23 modules. In the final section we state three structural relations as conditional propositions and discuss their potential physical interpretation as an open question. We make no physical claims.

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1 Introduction

1.1 Motivation

Formal verification of algebraic structures has advanced significantly with proof assistants such as Lean 4 and its mathematical library Mathlib [1, 2]. However, most formalized systems start from well-established mathematical foundations (groups, rings, categories) rather than from first principles of composition.

This paper takes a different approach: we start from the *minimal* axioms needed for a composition operation with dependency tracking, and ask what structural consequences are *forced* by these axioms alone.

1.2 Contributions

1. **Core Collapse** (Theorem 3): Four axioms force all dependency lists to be empty.
2. **Minimality** (Section 4): Associativity is redundant; exact additivity and no-duplicate are both necessary; the minimal system has three axioms.
3. **Finite-Free Dichotomy** (Theorem 13): The two-aspect dichotomy holds without finiteness assumptions on the type of rules.
4. **Unique Prime Basis** (Theorem 18): The basis $\{2, 3, 5, 7\}$ is uniquely determined by exclusion from the prime factors of three finite simple groups.
5. **Structural Relations** (Section 8): Three conditional structural relations, stated as open problems for physical interpretation.

1.3 What this paper does *not* claim

This paper is purely mathematical. We make no claims about physics, physical constants, or the structure of the universe. Section 8 states three structural relations as conditional propositions; the physical interpretation is explicitly left as an open question.

2 The Axiom System

Definition 1 (Causal Composition System). *A causal composition system consists of:*

- *A type C of rules;*
- *A type M of relata;*
- *A function $\text{input} : C \rightarrow \text{List } M$;*

- A binary operation $\circ : C \rightarrow C \rightarrow C$;

satisfying:

$$(A1) \quad \text{input}(\alpha \circ \beta) = \text{input } \alpha ++ \text{input } \beta \quad (1)$$

$$(A2) \quad (\text{input } \alpha).\text{Nodup} \quad (2)$$

$$(A3) \quad \circ \text{ is a total binary operation} \quad (3)$$

$$(A4) \quad (\alpha \circ \beta) \circ \gamma = \alpha \circ (\beta \circ \gamma) \quad (4)$$

Remark 2. The Lean formalization groups these into a type class *CompilerAxiom* with five fields: *compose*, *assoc*, *input*, *compose_input*, *input_nodup*. Axioms A1–A4 above correspond to the four non-trivial constraints. Section 4 proves that A4 (associativity) is redundant for Core Collapse, yielding a minimal system of three axioms (A1–A3).

3 Core Collapse

Theorem 3 (Core Collapse). *Let $(C, M, \text{input}, \circ)$ be a causal composition system satisfying A1–A4. Then for every $\alpha : C$,*

$$\text{input } \alpha = [].$$

Proof. Consider $\alpha \circ \alpha$. By A1,

$$\text{input}(\alpha \circ \alpha) = \text{input } \alpha ++ \text{input } \alpha.$$

By A2, this concatenated list has no duplicates.

Suppose $\text{input } \alpha = y :: t$ for some $y \in M$ and $t \in \text{List } M$. Then

$$(y :: t) ++ (y :: t) = y :: (t ++ (y :: t)).$$

The element y occurs at the head of the list, and again within the tail $t ++ (y :: t)$ (specifically, at the position where the second copy of $y :: t$ begins). This contradicts the Nodup property (A2).

Hence $\text{input } \alpha$ cannot be non-empty; it must be $[]$. $\square$

Corollary 4. *No rule has any dependency: $\forall \alpha : C, \forall x : M, x \notin \text{input } \alpha$.*

Remark 5. The Lean proof (*CoreCollapse.lean*, theorem *no_causal_input*) formalizes exactly this argument, using the *Nodup* property on lists.

4 Minimality

4.1 Associativity is redundant

Theorem 6. *Core Collapse holds in the system obtained by dropping A4 (associativity), provided A1–A3 are retained. Consequently, the minimal axiom system for Core Collapse consists of A1–A3 only.*

Proof. The proof of Theorem 3 only uses $\alpha \circ \alpha$ (composition of a rule with itself), never three distinct rules composed in sequence. Hence A4 (associativity for three-way compositions) is never invoked. The same argument works verbatim in the system without A4. $\square$

Remark 7. In Lean, this is established by defining a separate type class *MinimalCompilerAxiom* without the *assoc* field and re-proving Core Collapse (*AxiomIndependence.lean*, theorem *minimal_core_collapse*). The proof is identical.

4.2 Exact additivity is necessary

Theorem 8. *If A1 is weakened to sub-additivity $\text{input}(\alpha \circ \beta) \subseteq \text{input } \alpha \cup \text{input } \beta$, then Core Collapse fails.*

Proof. Counterexample: take $M = \text{Unit}$, $C = \text{Bool}$, $\circ \equiv \text{false}$, $\text{input true} = [\text{unit}]$. Sub-additivity and Nodup both hold trivially, but $\text{input true} \neq []$. $\square$

4.3 No-duplicate is necessary

Theorem 9. *If A2 is dropped, Core Collapse fails.*

Proof. The contradiction in Theorem 3 comes directly from applying Nodup (A2) to $\text{input } \alpha ++ \text{input } \alpha$. Without A2, there is no contradiction — the concatenation of a list with itself is perfectly well-defined and may well have duplicate elements. $\square$

5 Finite-Free Two-Aspect Dichotomy

Definition 10 (Amplitude). *An amplitude is a function $a : C \rightarrow \mathbb{C}$ with $|a(\alpha)|^2 = 1$ for all α and $a(\alpha \circ \beta) = a(\alpha)a(\beta)$.*

Definition 11 (Output). *An output is a function $o : C \rightarrow M$ with $o(\alpha \circ \beta) = o(\beta)$.*

Lemma 12. *If a is injective, then $\circ$ is commutative.*

Proof. $a(\alpha \circ \beta) = a(\alpha)a(\beta) = a(\beta)a(\alpha) = a(\beta \circ \alpha)$. Since a is injective, $\alpha \circ \beta = \beta \circ \alpha$. $\square$

Theorem 13 (Finite-Free Two-Aspect Dichotomy). *Given an amplitude a and an output o , we have:*

$$(\forall \alpha, \beta : C, o(\alpha) = o(\beta)) \vee \neg \text{Injective}(a).$$

No finiteness assumption on C is required.

Proof. Suppose a is injective. By Lemma 12, $\circ$ is commutative. Then

$$o(\alpha) = o(\beta \circ \alpha) = o(\alpha \circ \beta) = o(\beta),$$

where the first equality uses the output property with $\text{id} = \beta$, and the second uses commutativity. Hence o is constant on all of C . $\square$

Remark 14. *The Lean formalization (`TwoAspectNoFinite.lean`, theorem `two_aspect_dichotomy_no_finite`) proves this without any `[Finite C]` assumption, distinguishing it from earlier versions that required finiteness.*

6 Sequence Structure and Unique Basis

Definition 15 (The Sequence). *Let $s : \mathbb{N} \rightarrow \mathbb{N}$ be defined by $s_0 = 2$, $s_{n+1} = 3 + 2n$. The first few terms are:*

$$2, 3, 5, 7, 9, 11, 13, 15, \dots$$

Proposition 16. *The first four terms $\{s_0, s_1, s_2, s_3\} = \{2, 3, 5, 7\}$ are all prime. The fifth term $s_4 = 9 = 3^2$ is composite.*

Definition 17 (Prime Basis). *Let*

$$\text{MUST_HAVE} = \text{PrimeFactors}(12) \cup \text{PrimeFactors}(60) \cup \text{PrimeFactors}(168),$$

the union of prime factors of the orders of A_4 , A_5 , and $\text{PSL}(2, 7)$.

Theorem 18 (Unique Basis). $\text{MUST_HAVE} = \{2, 3, 5, 7\}$. *Moreover, if B is any set of primes with $\text{MUST_HAVE} \subseteq B \subseteq \text{MUST_HAVE}$, then $B = \text{MUST_HAVE}$.*

Proof. $|A_4| = 12 = 2^2 \cdot 3$, so $\{2, 3\} \subseteq \text{MUST_HAVE}$. $|A_5| = 60 = 2^2 \cdot 3 \cdot 5$, so $5 \in \text{MUST_HAVE}$. $|\text{PSL}(2, 7)| = 168 = 2^3 \cdot 3 \cdot 7$, so $7 \in \text{MUST_HAVE}$. The union is $\{2, 3, 5, 7\}$. $\square$

Theorem 19 (Necessity of Each Prime). *Removing any element of $\{2, 3, 5, 7\}$ breaks the coverage of at least one group:*

- *Removing 2 breaks A_4 : no group of even order can be represented without the prime 2.*
- *Removing 3 breaks all three groups.*
- *Removing 5 breaks A_5 .*
- *Removing 7 breaks $\text{PSL}(2, 7)$.*

Remark 20. *In Lean (`SequenceStructure.lean`), each of these four claims is a separate theorem: `removing_2_breaks_A4`, `removing_3_breaks_all`, `removing_5_breaks_A5`, `removing_7_breaks_PSL`. The `MUST_HAVE` set is characterized as `MUST_HAVE_eq_base` and the uniqueness is proved by `unique_base_is_2357`.*

7 Formalization

All results are formalized in Lean 4 with Mathlib. The repository contains 23 Lean modules. The build consists of 2089 jobs, compiles with 0 errors, and contains no `sorry`.

The repository is available at: <https://github.com/New-Beginning-Universe-Research-Group/CSQIT>

Key modules:

- `CompilerAxioms.lean`: axioms A1–A4 (the `CompilerAxiom` type class)
- `CoreCollapse.lean`: Theorem 3 (`core_collapse_summary`, `no_causal_input`)
- `AxiomIndependence.lean`: Theorems 6, 8, 9 (`minimal_core_collapse`)
- `TwoAspectNoFinite.lean`: Theorem 13 (`two_aspect_dichotomy_no_finite`)
- `SequenceStructure.lean`: Theorems 18, 19 (`MUST_HAVE_eq_base`, `unique_base_is_2357`, `removing_2/3/5/7_breaks_*`)
- `ObserverLayering.lean`: $\cos \theta = 0$ level structure (`cos_at_alpha_level`, `observed_equals_bare_at_alpha`)
- `QuantumCorrection.lean`: Symmetric polynomial gap (`e1_sq_sub_e3_eq_p1p2p4`, `Omega_sum_obs_eq_one`)

8 Structural Relations and Potential Physical Applications

The following three relations are Lean-verified theorems. Each is stated as a *conditional* proposition. The physical interpretation is left entirely as an open question; this paper does not claim that the physical universe satisfies any of the “if” conditions.

Proposition 21 (Prime Basis Forcing). *The prime factors of the orders of A_4 , A_5 , and $\text{PSL}(2, 7)$ have union $\{2, 3, 5, 7\}$. Any theory whose symmetry group is among these three groups must use the prime basis $\{2, 3, 5, 7\}$.*

Proposition 22 (Symmetric Polynomial Gap). *Let $e_1 = 2 + 3 + 5 + 7 = 17$ and $e_3 = 2 \cdot 3 \cdot 5 + 2 \cdot 3 \cdot 7 + 2 \cdot 5 \cdot 7 + 3 \cdot 5 \cdot 7 = 247$ denote the first and third elementary symmetric polynomials of $\{2, 3, 5, 7\}$, let $e_4 = 2 \cdot 3 \cdot 5 \cdot 7 = 210$, and let $N = 2e_4$. Then:*

$$e_1^2 - e_3 = 17^2 - 247 = 289 - 247 = 42 = 2 \cdot 3 \cdot 7 = p_1 p_2 p_4,$$

and $N = 2 \cdot 210 = 420$.

Remark 23. *Proposition 22 is an algebraic identity, provable by direct computation. It depends on no physical assumptions. In Lean (`QuantumCorrection.lean`), this is `e1_sq_sub_e3_eq_p1p2p4`.*

Proposition 24 (Level $\cos \theta = 0$). *Define $n = N/4 = 420/4 = 105$. Let $\theta = 2\pi n/N = 2\pi \cdot 105/420 = \pi/2$. Then $\cos \theta = 0$. If a physical quantity q at level n has a correction factor of the form $1 + \Delta \cos \theta$ for some Δ , then at level $n = 105$ the correction factor equals 1 (the correction vanishes).*

Remark 25. *Propositions 21–24 are mathematical. They do not assert that A_4 , A_5 , $\text{PSL}(2, 7)$ are the symmetry groups of nature, nor that the physical universe has a discrete level structure. Whether the physical universe realizes these structures is an empirical question, not addressed here.*

9 Discussion

9.1 Open Mathematical Questions

1. Is there a classification-theoretic reason for the particular triple A_4 , A_5 , $\text{PSL}(2, 7)$? These are three of the smallest non-abelian finite simple groups. Does this triple have a unified characterization?
2. Can the exclusion argument be extended to other families of finite simple groups? What prime bases do they force?
3. Is there a general characterization of sequences $s : \mathbb{N} \rightarrow \mathbb{N}$ whose first k terms are exactly the prime factors of a given set of finite simple groups?
4. Proposition 22 gives $e_1^2 - e_3 = p_1 p_2 p_4$. Are there similar gap identities for other sets of primes?

9.2 Open Physical Questions

Whether the structural relations of Section 8 have physical content is an empirical question. In particular:

- Is the symmetry group of nature (if any) among A_4 , A_5 , $\text{PSL}(2, 7)$?

- Does the physical universe have a discrete level structure with $n = 105$ playing a distinguished role (the level at which $\cos \theta = 0$)?
- If the symmetric polynomial gap $e_1^2 - e_3 = 42$ has physical meaning, does it correspond to the magnitude of a quantum correction?

These questions are outside the scope of this paper.

9.3 Scope and Limitations

This paper is purely mathematical. No physical claims are made. Numerical coincidences observed in adjacent work are not endorsed here. The three structural relations of Section 8 are stated as conditional propositions; their physical relevance, if any, remains an open question.

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References

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